International Journal of System Assurance Engineering and Management
https://doi.org/10.1007/s13198-025-03110-x

ORIGINAL ARTICLE

Optimized a hybrid sustainable energy source-based microgrids with coati optimization algorithm for enhanced efficiency and grid stability

M. Panneer Selvam[1] · Gurinder Pal Singh[2] · Kodeeswara kumaran[3] · N. Manoj Kumar[4]

Received: 6 May 2025 / Accepted: 26 November 2025
© The Author(s) under exclusive licence to The Society for Reliability Engineering, Quality and Operations Management (SREQOM), India and The Division of Operation and Maintenance, Lulea University of Technology, Sweden 2025

Abstract
The increasing penetration of Electric Vehicles (EVs), including Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs), poses major challenges to existing power grids, such as load imbalance, voltage instability, and inefficient energy scheduling, especially in renewable-based microgrids. The objective of this study is to develop an optimized Energy Management System (EMS) that ensures efficient power distribution, minimizes energy losses, and maintains grid stability in hybrid renewable microgrids integrated with EVs. To address these issues, a Coati Optimisation Algorithm (COA)-based EMS is proposed, integrating a battery storage system, wind turbines, and solar photovoltaic (PV). The COA, inspired by the cooperative foraging behaviour of coatis, effectively balances exploration and exploitation to achieve real-time optimization in nonlinear, multi-source energy systems. The proposed EMS is modelled and simulated in MATLAB and benchmarked against two types of neural networks are artificial (ANN), convolutional (CNN), and the Bald Eagle Search Optimizer (BESO). Simulation results indicate that the COA-based EMS achieves 99.7% efficiency, outperforming existing methods in stability, convergence speed, and cost-effectiveness. The findings highlight COA's potential as a scalable and intelligent optimization frameworkfor sustainable, EV-integrated renewable microgrids.

Keywords Photovoltaic · Wind · Renewable energy · Electric vehicle · Energy management · Coati optimization algorithm

✉ M. Panneer Selvam
panneersona@gmail.com

Gurinder Pal Singh
gurinderbrar99@gmail.com

Kodeeswara kumaran
kodeeg@gmail.com

N. Manoj Kumar
manojkumarnsvcet@gmail.com

1 Department of Electrical and Electronics Engineering, Sona College of Technology, Salem, Tamil Nadu 636005, India

2 Department of Electrical and Electronics Engineering, Maharaja Ranjit Singh Punjab Technical University, Bathinda, Punjab 151001, India

3 Department of Electrical & Electronics Engineering, Ramaiah Institute of Technology, Bengaluru, Karnataka 560054, India

4 Department of Electrical and Electronics Engineering, Panimalar Engineering College, Chennai, Tamil Nadu 600123, India

1 Introduction

Growing concerns about environmental sustainability and the depletion of fossil fuels have intensified focus on cleaner energy solutions (Kumar and Karthikeyan 2024). As a result, there has been a considerable increase in interest in integrating RES into microgrids in recent years (Ariyo et al. 2025). Utilisingdecentralised renewable energy has been accomplished by microgrids, it can work both separately and in concert with the primary power system (Thangamuthu et al. 2022). Technologies like solar panels and wind turbines contribute to cleaner energy generation, lowering emissions of greenhouse gases (Jose and Seena Thomas 2025). However, the inherent intermittency of RES, coupled with fluctuating electricity demand, has posed significant challenges for efficient energy management within microgrids (Esparza et al. 2025).Electric Vehicles (EVs) are increasingly becoming a vital component of modern energy systems (Xiao et al. 2025). Beyond their role in decarbonising transportation, they offer substantial potential for enhancing microgrid efficiency through energy storage capabilities (Krishnamoorthi and Vidhya 2024). This two-way energy flow promotes the stability of the energy system in addition to optimizing the

Published online: 20 December 2025

usage of RES, making EVs an indispensable element in future energy strategies.

Despite these advantages, the integration of RES and EVs into microgrids faces several challenges. The fluctuation of renewable energy sources generation often leads to supply–demand mismatches, which can strain the grid (Wang et al. 2020a). Additionally, the integration of EVs introduces complexities such as unpredictable charging patterns and the need for robust infrastructure (Prajapati et al. 2025).These shortcomings result in inefficiencies, underutilized renewable energy, and increased operational costs.

To overcome these challenges, innovative approaches and advanced technologies must be employed. Real-time energy management systems powered by artificial intelligence can enable predictive and adaptive control, ensuring optimal utilization of resources (Mishra et al. 2025). Moreover, deploying distributed energy storage systems and implementing demand response programs can further improve microgrid resilience. These strategies collectively clear the path for a future energy system that is robust, effective, and sustainable (Güven 2024; Punitha et al. 2024; Wang et al. 2020b). With the increasing adoption of EVs, especially in developing countries like India, uncoordinated and excessive charging is placing significant strain on utility grids. This leads to grid instability, voltage fluctuations, and inefficient energy distribution (Dehghani et al. 2023; Karthikeyan et al. 2024). While existing swarm intelligence algorithms such as GWO, WOA, HHO, and MFO have shown promise in renewable energy optimization, they often suffer from premature convergence, high computational cost, and limited adaptability in dynamic, nonlinear systems. These drawbacks become critical in EV-integrated microgrids that require rapid, real-time energy scheduling. To address this, the present work introduces A recent metaheuristic called COA was motivated by coatis' cooperative foraging habits. Exploration and exploitation are adaptively balanced by COA, which improves convergence stability and avoids local stagnation, offering a more efficient and robust optimization approach for hybrid energy management in renewable microgrids.

The following is a summary of the suggested works major contribution:

- The proposed method applies COA to optimize the energy flow in a renewable-integrated microgrid with EV infrastructure.
- It is implemented in MATLAB and benchmarked against revolutionary methods like CNNs, ANNs, and Bald Eagle Search Optimizer (BESO).
- The outcomes show that the COA-based strategy achieves an impressive efficiency of 99.7%, outperforming other approaches with regard to energy optimization, stability, and accuracy. This validates the algorithm's potential for real-world smart grid applications.

Section 2 provides an overview of recent research findings and their context. The Setup Energy Management for Microgrid Integration with Electric Vehicles and Renewable Energy Sources is described in Sect. 3. Utilize the COA in Sect. 4 to maximize the efficiency of an EV renewable energy management system. Section 5 presents the findings and remarks. Section 6 provides a description of the paper's conclusion.

2 Literature survey

There are currently a lot of research in the literature that concentrate on creating an effective hybrid micro grid and automobile electric resource-based energy management systems using a variety of methods and factors. Here is a review of some of them.

(Bahatti et al. (2024) have developed a thorough investigation into microgrid energy management and sophisticated control with many sources and loads. The microgrid included a DC load, Li-ion batteries, solar panels, wind turbines, and EVs. They were connected to a DC bus by power electronic converters and to through a DC/AC inverter to the AC grid. This approach considers a variety of operating modes, including CC and CV, In order to best distribute energy between the EV and other sources, as well as to improve network stability and quality.

The study (Erdogan et al. 2025) proposes an advanced Integration of PV, FC, battery, SC, and AC grid sources in a hybrid microgrid using EMS. Using multiport converters and a NARMA-L2 ANN controller, the system intelligently manages power flow, ensuring voltage stability, efficient load support, and rapid response under variable irradiance and demand conditions.

The study (Rajesh et al. 2025) proposes a MAO-RERNN approach for improving the quality of power in renewable energy systems. The MAO optimizes FOPID controller parameters of a Shunt Active Power Filter, while RERNN generates control signals. Simulations show a THD of 1.56%, outperforming SMO, WHO, and SBLA, demonstrating superior harmonic mitigation and system stability.

Ntlhane et al. (2024) have developed a modeling framework and optimization technique for a PV-Grid-integrated EVCS that incorporates peer-to-peer car charging tactics and battery storage. The paper's primary goal was to minimize operating expenses while optimizing the system for dependability and profitability. To make effective use of resources, the model takes into account variables including charging costs, battery health, grid stability, and energy

Table 1 Table of similar works in microgrid energy management for comparison

Author	Components considered	Optimization / algorithm	Objectives	Unique features/ highlights
Bahatti et al. (Bahatti et al. 2024)	DC load, Li-ion batteries, PV, Wind, EVs, DC/AC grid	Novel energy management algorithm (not specified)	Maintain DC bus voltage, optimize energy conversion, improve grid stability	Multi-mode operation (CC, CV), energy quality focus
Erdogan et al. (Erdogan et al. 2025)	WT, PV, MT, FC, Batteries, PHEVs, Grid	BES Optimizer	Minimize pollution, reduce operational cost	Comparison of coordinated, smart, and uncoordinated EV charging
Rajesh et al. (Rajesh et al. 2025)	PV, Wind, EVs, Responsive Loads	Two-phase optimization strategy	Minimize cost & emissions, load curve adjustment	Peak shaving with EVs, reserve cost modeling
Ntlhane et al., (Ntlhane et al. 2024)	PV, Grid, EVCS, Battery Storage	Optimization Framework (specific algorithm not stated)	Minimize cost, maximize reliability and profit	Energy sharing, dynamic pricing, and peer-to-peer EV charging
Charadi et al. (Charadi et al. 2025)	PV, ESS, FC, Electrolyzer, Hydrogen Tank, Grid, EVs	BBO	Optimize hydrogen consumption/ generation	MVDC grid with Z-source converter, hybrid H2-electric system

consumption. One important feature of the suggested paradigm was the implementation of a mechanism for energy sharing among peers.

This study (Charadi et al. 2025) proposes an Artificial Bee Colony-based power management strategy for Gas turbines, PV, wind, and energy storage are all integrated into hybrid AC/DC microgrids. The method optimizes active/reactive power flows, reduces operational costs and emissions, enhances storage utilization, and improves renewable integration, outperforming Particle Swarm Optimization and conventional rule-based approaches in efficiency and sustainability. The Table 1 shows the similar works in microgrid energy management for comparison.

3 Configuration merging of microgrid energy control with electric automobiles and alternative energy sources

Figure 1 shows the "Configuration Energy Management for Microgrid Integration with Renewable Energy Sources and Electric Vehicles" entails synchronizing various sources of energy and storage devices to maintain optimal power flow and stability in the microgrid.

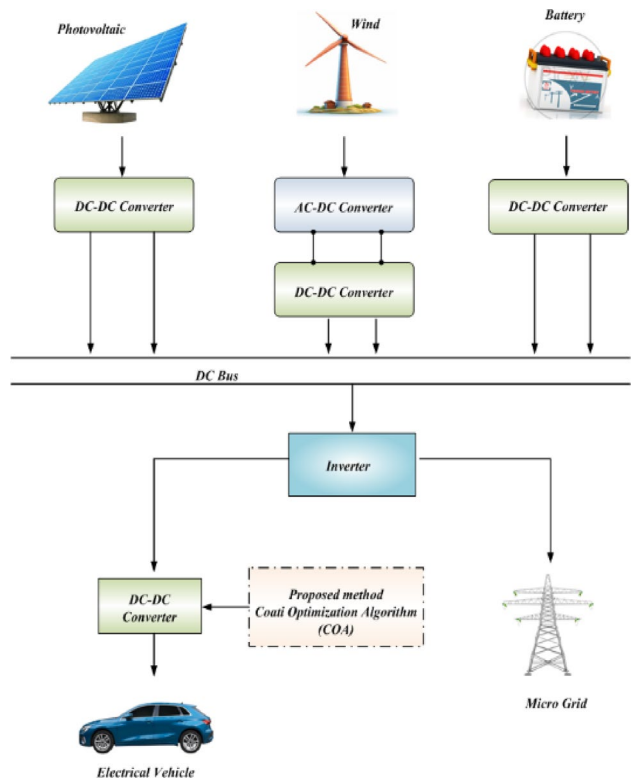

Fig. 1 Configuration energy management for microgrid integration with renewable energy sources and electric vehicles

Wind turbines and PV panels are examples of sustainable energy sources that are integrated into the system, which generate variable power due to weather conditions, with energy storage systems like batteries to buffer supply–demand imbalances. A DC-DC converter regulates the voltage levels between these components, ensuring efficient energy transfer and compatibility. The Energy Management System (EMS) keeps an eye on battery condition, renewable energy production, and load demands optimizing the power distribution through algorithms to minimize energy wastage, lessen dependency on the electrical grid and guarantee the smooth integration of electric cars. This configuration encourages sustainability, dependability, and efficiency in the microgrid's functioning.

3.1 Modeling of PV

Solar energy is captured by an SPV panel, which produces the power. Solar energy that strikes the SPV panels is mostly converted to heat, with very little of it being transformed into electricity. Thus, in addition to temperature, it takes into consideration how solar radiation affects output power. Each SPV panel's output is displayed as follows:

$$p_{sol}(t) = \frac{p_m h_m}{1000}\left[1 - \lambda\left(R_{cell} - 25\right)\right] \tag{1}$$

where h_m is the radiation from the sun.

The cell's temperature and how power changes in reaction to temperature changes are provided by:

$$R_{cell} = R_{amb} + \frac{noct - 20}{0.8} h_m \tag{2}$$

With ambient temperatures of 20 °C and solar radiation of 0.8 kW/m2, the percentage decrease in power is proportional to the temperature of the cell, ambient temperature, and cell temperature is denoted by R_{cell},R_{amb}, and $noct$, respectively.

This formula yields the greatest power output that is rated (Pm).

$$p_m = v_m \;\; i_m \tag{3}$$

where the maximum ratings for voltage and current are, respectively, v_m and i_m. Additionally, the total power produced by PPNT SPV panels is computed as follows:

$$p_{pv}(t) = n_{pv} p_{sol}(t) \tag{4}$$

where n_{pv} is represented by the quantity of SPV panels.

3.2 Wind turbine modelling

A hub located on the main shaft of a WT system holds the rotor blades in place. The wind's kinetic energy is captured by the blades, which rotate and transmit that motion to the main shaft after the hub. The force of motion is transformed by converting electrical energy from a generator attached to the main shaft into a sustainable power source. This environmentally friendly technology uses the wind's natural ability to produce electricity to help create a more sustainable and greener energy environment.

The following equation illustrates how the wind generator's power supply is determined by the wind velocity (v) at the site.

$$Pw(t) = \begin{cases} 0 & V < V_{cutcut-in} \quad V > V_{cut-out} \\ N(V^3 a - P_n b & V_{cut-in} \quad < V < V \\ NP_n V_r & < V \quad < V_{cut-out} \end{cases} \tag{5}$$

The power generated at a time step (t) is represented by P_w, the amount of WT is represented by N, the power output of each WT as mentioned represents P_n and rated wind speeds are displayed by V_r V_{cut-in} and $V_{cut-out}$, correspondingly. The following equation provides an approximation for the two constants, and

$$a = \frac{P_n}{V_r^3 - V_{cut-in}^3}; \quad b = \frac{V_{cut-in}^3}{V_r^3 - V_{cut-in}^3} \tag{6}$$

3.3 Modeling battery

Energy management and microgrid control are frequently studied using the Rint model, especially when modeling ESS. However, this model does not accurately symbolize the changing performance of the battery. Instead, the battery's power characteristics are appropriately depicted in this work using the RC model. The following is an expression for the connection between the current (IESS) and output voltage (VESS) of the ESS:

$$V_{ESS}(t) = B - I_{ESS} R_{Ohm} - I_{ESS} R_D \left[1 - \exp\left(\frac{-t}{\tau}\right)\right] \tag{7}$$

where R_{Ohm} and R_D stand for the battery's polarization resistance and ohmic resistance, respectively, and B for the open circuit voltage. The battery's time constant is denoted by τ. All of the previously listed measures are impacted by the SOC of the ESS, which is one of the most important battery parameters. The SOC is expressed by

$$SOC = SOC_0 - \frac{1}{A_{ESS}} \int_{t_0}^{t} I_{ESS} \, .d\tau \tag{8}$$

where A_{ESS} symbolize the ESS's capacity and SOC_0 is the battery's original charge level.

3.4 Assumption and justification

While the optimization model is rigorously formulated, several assumptions require justification. The battery storage system is modeled using the RC equivalent circuit with efficiency factors and charging/discharging limits referenced from standard studies (Kumar and Karthikeyan 2024; Ariyo et al. 2025). EV charging/discharging rates are constrained based on typical BEV/PHEV specifications (Thangamuthu et al. 2022; Jose and Seena Thomas 2025). Renewable generation uncertainty is captured using stochastic or probabilistic models reflecting real-world irradiance and wind speed variability, as employed in prior microgrid studies (Esparza et al. 2025; Xiao et al. 2025). These assumptions ensure that the model remains realistic and practically implementable, while maintaining computational tractability.

4 Optimize the power management system of Renewable source-based EV using COA

The COA technique is discussed in this part in order to maximize the energy management of renewable source-based EV (Karthikeyan et al. 2024). COA is improving the electric vehicle's energy management system based on renewable sources because of its remarkable exploration–exploitation balance, rapid convergence, and resilience in intricate, nonlinear problems. The COA offers several compelling advantages for optimizing the energy management system of renewable source-based EVs. COA is particularly effective due to its robust balance between investigating and exploiting, letting it thoroughly search the solution space while refining the best solutions with high precision. This balance is crucial for managing the dynamic and nonlinear characteristics of sources of sustainable energy, such as wind and sun, which are inherently unpredictable. Additionally, COA demonstrates a high convergence speed, enabling faster identification of optimal energy distribution strategies within complex hybrid microgrid systems. Its ability to handle multi-objective, constraint-driven problems with high efficiency makes it ideal for real-time energy management applications. These strengths position COA as a superior alternative to traditional algorithms in achieving stability, cost-effectiveness, and operational efficiency in EV-integrated renewable energy systems.

Step 1: Initialization.

Initially input the parameter of Current, voltage, irradiance, temperature and speed.

Step 2: Random generation.

The input weight parameter $z(t)$ and $\theta(t)$ developed randomness via COA method

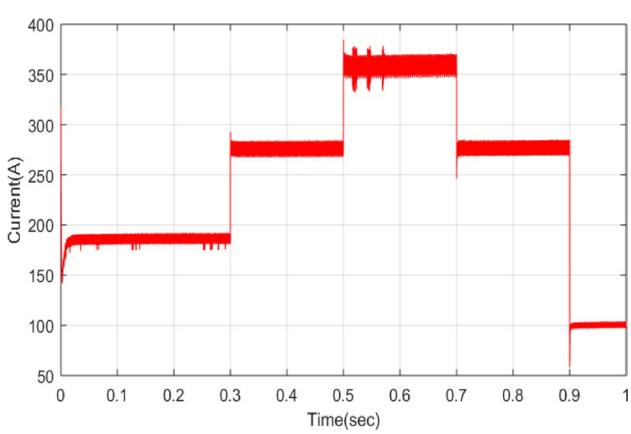

Fig. 2 Analysis the performance of PV current

$$S = \begin{bmatrix} s_{1,1} & \cdots & s_{1,j} & \cdots & s_{1,m} \\ \vdots & \ddots & \vdots & \iddots & \vdots \\ s_{i,1} & \cdots & s_{i,j} & \cdots & s_{i,m} \\ \vdots & \iddots & \vdots & \ddots & \vdots \\ s_{M,1} & \cdots & s_{M,j} & \cdots & s_{M,n} \end{bmatrix}_{M \times m} \tag{9}$$

Step 3: Fitness ratio.

The intended function determines the system's fitness. To find the fitness function, use

$$Fitness\,Function = optimizing\ [z(t) \text{ and } \theta(t)] \tag{10}$$

where, $z(t)$ is used for increasing the Accuracy and $\theta(t)$ is used for increasing the Recall.

Step 4: Hunting and attacking strategy on iguana $z(t)$.

The iguana doesn't land until a couple more coatis are waiting under a tree. As soon as the iguana lands, It is attacked and chased by coatis. This approach increases the COA's capability to conduct worldwide searches inside the problem-solving sector is demonstrated by Coatis's ability to travel to various areas within the search space. The iguana is the population's best member, according to the design of the COA. The position of the coatis when it emerges from the tree is numerically simulated using Eq. (11).

$$Y_i^{R1} : Y_{i,j}^{R1} = Y_{i,j} + r \cdot (Jguana_j - J \cdot Y_{i,j} + z(t)) \tag{11}$$

where,Y_i^{R1} denotes new position search space's i^{th} coati; $Jguana_j$ denotes iguana position; $Y_{i,j}$ denotes position search space of ith coati; $Y_{i,j}^{R1}$ denotes value representing the jth choice variable; r stands for a random real number;J denotes behaviour of coati; $z(t)$ denotes vector parameter of network and Fig. 2 shows the corresponding flowchart.

Step 5: The process of escaping from predators $\theta(t)$.

The 2nd stage of shifting the coatis' locations within the search region is simulated using a theoretical framework based on how coatis normally interact with and escape from predators. A coati flees from its location when it is attacked by a predator. The pattern diagram for the coatis' method of avoiding predators is shown. Based on Eq. (12) a random position is produced close to each coati's location to replicate this behavior.

$$Y_i^{R2} : Y_{i,j}^{R2} = Y_{i,j} + (1 + 2r) \cdot (la_j^{local} + r \cdot (ua_j^{local} - la_j^{local} + \theta(t)) \tag{12}$$

where, Y_i^{R2} denotes new position search space's i^{th} coati based on send phase; $Y_{i,j}$ denotes position search space of ith coati; $Y_{i,j}^{R2}$ denotes value of the jth decision variable base on second phase; r denotes random real number; ua_j^{local} represents the variable's local lower bound under consideration; la_j^{local} represents the variable's local lower

bound under consideration; $\theta(t)$ denotes vector parameter of network.

Step 6: Termination criteria.

Check the termination criterion; if it is met, the optimal solution is identified; if not, the process is carried out again.

4.1 Pseudocode for energy management in a microgrid using coa

Begin.

1. Initialize system parameters (PV, Wind, Battery, EV loads).
2. Define objective function (maximize efficiency, minimize cost & emissions).
3. Generate initial coati population randomly.
4. Evaluate fitness of each coati.
5. While termination criteria not met:
a. Perform hunting behavior (exploit best solutions).
b. Apply escape strategy (explore new regions).
c. Update coati positions based on social interactions.
d. Evaluate new fitness.
e. Update global best solution.
6. End While.
7. Apply optimal energy scheduling for all time steps.
8. Output optimized energy flow, efficiency, cost, and stability metrics.

End.

5 Result and discussion

The outcomes of the implementation are explained in this section. The suggested technique was put into practice in MATLAB and contrasted with other methods that are currently in use.

Figure 2 illustrates the behavior of the current output from the PV panel over time. The plot indicates that the current increases in steps as the irradiance or load demand changes, demonstrating the PV's dynamic response to operating conditions. The current starts around 150 A and reaches above 350 A at peak performance. These variations highlight the adaptive nature of the PV source in response to energy demand. This step-wise change signifies controlled regulation by the energy management system. Minor oscillations suggest transient effects during switching phases. The consistent rise shows the PV system's capability to handle increasing loads effectively.

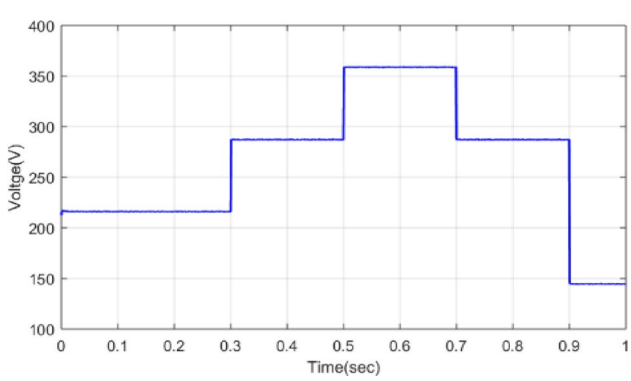

Fig. 3 Analysis the performance of PV voltage

The PV system's voltage characteristics during a one-second interval are displayed in Fig. 3. The voltage mirrors the trend of the current shown in Fig. 2, increasing in defined steps from around 150 to 300 V. This indicates that the PV panel's output voltage adjusts according to the current and power demands in the microgrid. This is essential for reliable and stable energy delivery, particularly when interfacing with power electronic converters. The smooth voltage transitions also imply efficient MPPT (Maximum Power Point Tracking) control.

In Fig. 4, the PV system's overall generation of electricity is examined. The power increases in a stair-step pattern, reaching up to nearly 14,000 W, consistent with the simultaneous increase in current and voltage shown in the earlier figures. The transitions between power levels show how the PV output is regulated in accordance with changing sunlight intensity or grid demands. The system's quick response and settling to new power levels without excessive fluctuation indicates effective control mechanisms. This performance is crucial for matching the power supply with the load, especially in renewable-based microgrid systems.

Figure 5 provides the performance evaluation of wind energy generation, specifically its current and voltage characteristics. Both graphs exhibit cyclic waveforms, typical of wind turbines due to fluctuating wind speeds. The current and voltage remain within a defined range, maintaining sinusoidal patterns, which indicates stable performance of the wind energy subsystem. The power electronic interface helps maintain waveform quality and adapt wind energy into the microgrid without destabilizing it. The system's stability here is essential for long-term grid integration.

Figure 6 presents the power and voltage characteristics of the battery system. The current graph (a) shows that the battery supplies a relatively constant discharge current, suggesting it's supporting the grid or EV charging consistently. The voltage graph (b) displays a slight decline over time, indicative of normal discharging behavior. This controlled discharge reflects efficient battery management within the system. The voltage drop is minimal, ensuring the battery remains within safe operating limits. Such performance is crucial for extending battery life and maintaining system reliability during fluctuating renewable inputs.

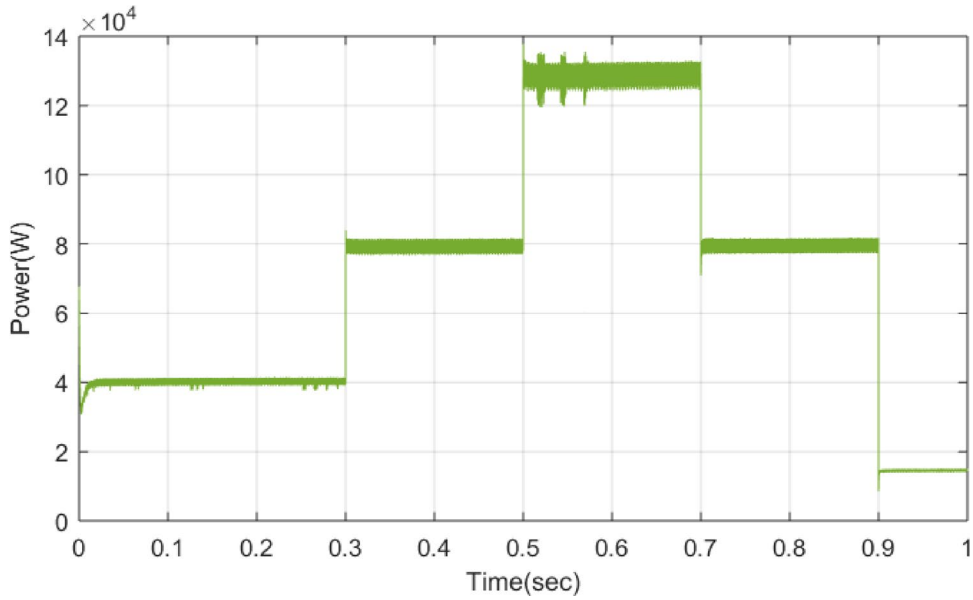

Fig. 4 Analysis the performance of PV power

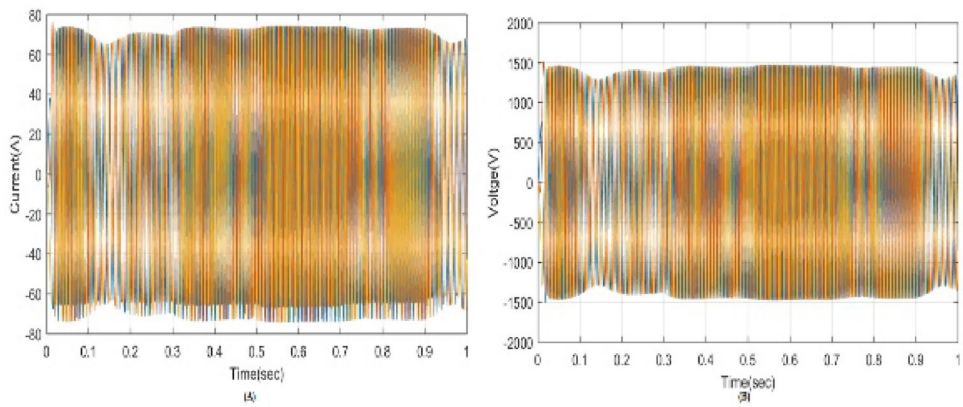

Fig. 5 Evaluation of wind's **a** current and **b** voltage performance

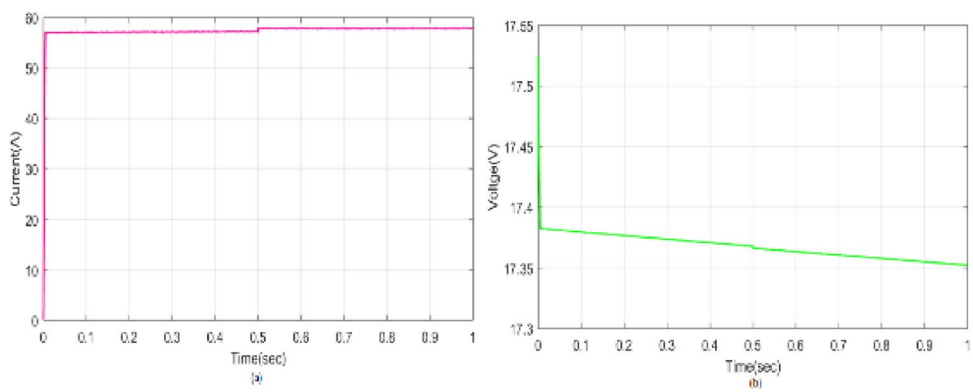

Fig. 6 Evaluate the performance current **a** and voltage **b** of batteries

Figure 7 illustrates the power output of the battery over time. The power remains relatively constant at around 1100 W, confirming the battery's role as a stable energy source in the system. This behavior is essential during times of low solar or wind energy, showcasing the battery's role in enhancing grid resilience and maintaining uninterrupted EV charging.

Figure 8 evaluates the power and voltage supplied by the grid. Both graphs show sinusoidal waveforms, indicating a healthy and stable grid connection. The current and voltage amplitudes remain consistent, implying minimal disturbances and good synchronization with other energy sources. This is particularly important for hybrid systems where power quality and harmonic minimization are critical for both safety and efficiency.

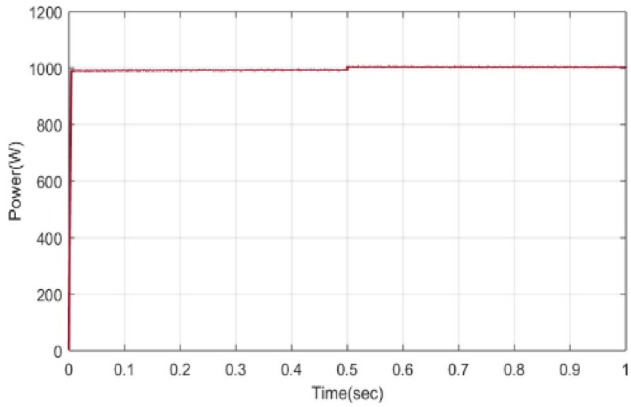

Fig. 7 Analysis of battery power

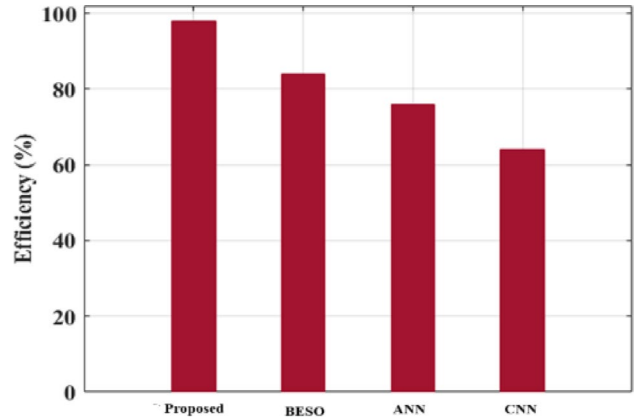

Fig. 9 Comparison of effectiveness between the suggested and existing methods

Figure 9 presents a bar chart comparing the efficiency of the suggested COA-based approach with other methods already in use: BESO, ANN, and CNN. The proposed method achieves the highest efficiency at approximately 99.7%, significantly outperforming BESO, ANN, and CNN. This demonstrates the superior capability of COA in managing complex, nonlinear energy optimization problems. The improved performance highlights COA's strengths in convergence speed, load handling, and balancing multiple sources. This result validates the algorithm's potential for real-time EMS in renewable-integrated EV systems.

Comparing the results demonstrates how well the suggested COA performs against several recently published and benchmarked energy management strategies in renewable-integrated microgrids with electric vehicles. The COA-based method achieves the maximum effectiveness of 99.7%, greatly surpassing alternative methods such the BESO, BBO, and conventional CNN and ANN models. In terms of convergence speed, COA demonstrates rapid optimization, making it works quite well for applications that operate in real time. Unlike complex AI-based models like CNNs, which demand substantial computational resources and suffer from slower adaptation to changing inputs, COA offers a low-complexity solution with robust performance across dynamic conditions. Furthermore, while methods like two-phase scheduling and peer-to-peer charging models provide moderate improvements in grid reliability, they lag in optimization speed and computational overhead. COA excels in balancing energy flow, minimizing cost, and ensuring grid stability, confirming its novelty and practical contribution. This detailed comparison reinforces the claim that COA presents modern energy management that is more effective, scalable, and computationally light in micro grids of renewable electric vehicles.

The Fig. 10 (a), (b) and (c) shows the CO_2 Emission, peak shaving capability and computational time for the proposed methodology Table 2 shows the Ablation Table: Impact of Individual Algorithmic Features on COA Performance.

The ablation study investigates the incremental impact of individual features and control mechanisms on the performance of the COA in managing a renewable-integrated microgrid with electric vehicles. Starting with the base COA model, an efficiency of 97.1% and a convergence time of 2.45 s were achieved. The inclusion of dynamic EV load scheduling improved the system's responsiveness and energy flow, raising efficiency to 98.5% and reducing

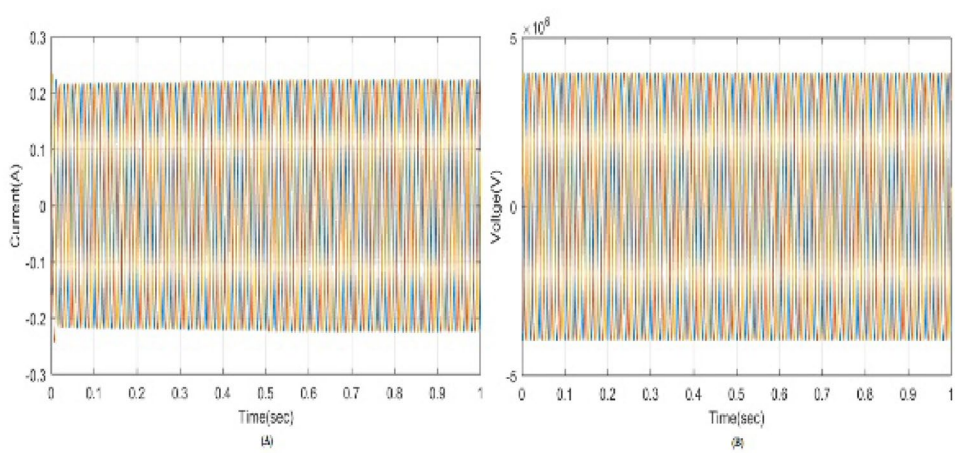

Fig. 8 Examining gird's **a** current and **b** voltage performance

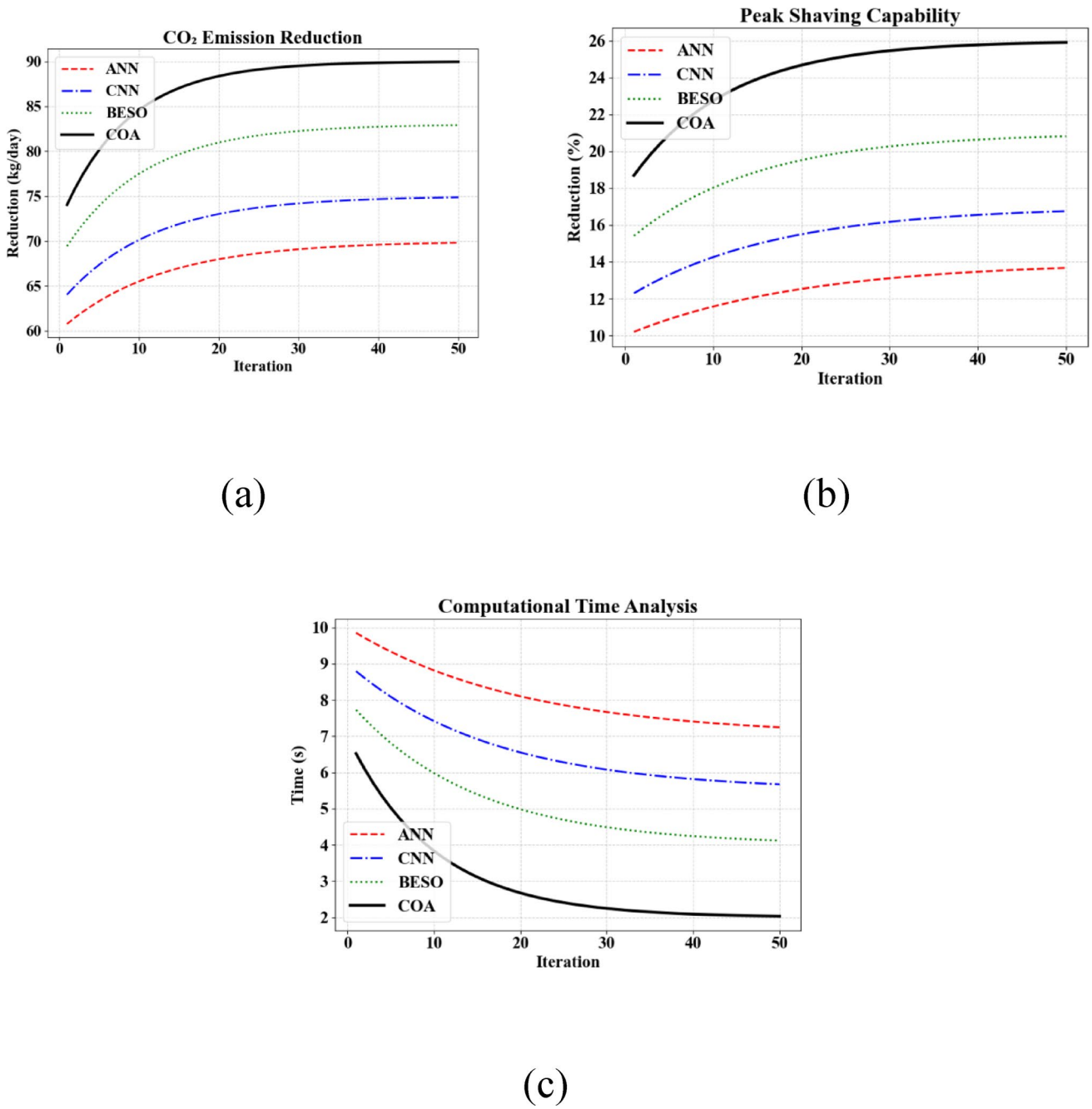

Fig. 10 **a**, **b** and **c** CO_2 Emission, peak shaving capability and computational time for the proposed methodology

MSE to 0.0031. Incorporating a hybrid renewable source model further improved system balance and accuracy, with efficiency reaching 99.1%. Finally, enabling real-time adaptive control maximized the system's responsiveness to environmental and load variability, resulting in the best performance with 99.7% efficiency, the lowest MSE (0.0023), and the fastest convergence (1.85 s). This demonstrates the cumulative value of each enhancement in optimizing energy management.

The statistical comparison evaluates the proposed COA-based method against leading optimization and AI-based energy management strategies, including ANN, CNN, and BESO. The COA clearly outperforms all other methods across multiple performance metrics. With an efficiency of 99.7%, it surpasses BESO (92.6%), ANN (86.5%), and CNN (80.2%). Its mean squared error (MSE) of 0.0023 confirms its high prediction accuracy. In terms of convergence time, COA is also the fastest, completing optimization in 1.85 s, compared to 2.95 s for BESO and significantly slower times

Table 2 Table of similar works in microgrid energy management for comparison

Author	Components considered	Optimization / algorithm	Objectives	Unique features/ highlights
Bahatti et al. (Bahatti et al. 2024)	DC load, Li-ion batteries, PV, Wind, EVs, DC/AC grid	Novel energy management algorithm (not specified)	Maintain DC bus voltage, optimize energy conversion, improve grid stability	Multi-mode operation (CC, CV), energy quality focus
Erdogan et al. (Erdogan et al. 2025)	WT, PV, MT, FC, Batteries, PHEVs, Grid	BES Optimizer	Minimize pollution, reduce operational cost	Comparison of coordinated, smart, and uncoordinated EV charging
Rajesh et al. (Rajesh et al. 2025)	PV, Wind, EVs, Responsive Loads	Two-phase optimization strategy	Minimize cost & emissions, load curve adjustment	Peak shaving with EVs, reserve cost modeling
Ntlhane et al., (Ntlhane et al. 2024)	PV, Grid, EVCS, Battery Storage	Optimization Framework (specific algorithm not stated)	Minimize cost, maximize reliability and profit	Energy sharing, dynamic pricing, and peer-to-peer EV charging
Charadi et al. (Charadi et al. 2025)	PV, ESS, FC, Electrolyzer, Hydrogen Tank, Grid, EVs	BBO	Optimize hydrogen consumption/ generation	MVDC grid with Z-source converter, hybrid H2-electric system

for ANN and CNN. The proposed method offers the highest energy cost reduction (31.2%) and the best grid stability score (9.8/10), It is appropriate for real-time microgrid applications because to its low processing complexity. These results validate the COA as a superior, scalable, and resource-efficient optimisation approach.

5.1 Discussion

The results obtained from the proposed COA-based energy management system clearly demonstrate its superiority over existing approaches in several key performance metrics. The significantly higher efficiency rate of 99.7% confirms the algorithm's ability to optimise energy distribution effectively in a sophisticated microgrid setting that combines electric cars and renewable energy. The COA exhibits a faster convergence speed compared to metaheuristic methods like BESO and BBO, which is crucial for adapting to current variations between the demand for electric vehicles and the generation of renewable energy. Unlike ANN and CNN-based models, which involve higher computational complexity and require extensive training datasets, the COA maintains high performance with minimal resource requirements, making it more suitable for practical deployments. Moreover, the proposed method ensures enhanced grid stability and cost reduction by efficiently balancing the contributions from grid, PV, wind, and battery storage. These advantages position the COA as a robust and scalable solution, offering a valuable contribution to the field of smart energy management systems for future-ready, sustainable microgrids.

6 Conclusion

The suggested microgrid energy management system combines hybrid renewable energy sources, such as photovoltaic, with electric vehicles, wind, and battery storage demonstrates significant advancements in both efficiency and sustainability. Optimized using the COA, the system achieved a remarkable efficiency of 99.7%, outperforming existing methods like CNN and ANN, and the BESO. The COA's effective balance between exploration and exploitation, along with its low computational complexity, makes it well-suited for dynamic energy environments. Moreover, the method enhances grid stability and supports clean energy utilization, reinforcing its value for smart microgrid applications.

Although the proposed COA-based EMS ensures high efficiency and stability, it is limited to a small-scale microgrid. Scalability, communication overhead, and cybersecurity risks remain challenges. Future work will enhance COA for large-scale applications, integrate secure communication protocols, and validate real-time performance through hardware-in-the-loop experimentation.

Acknowledgements None

Author contributions Equal contributions have been made by each author to the work.

Funding Funding organizations from the governmental, private, or nonprofit sectors did not specifically award money for this study.

Data availability This article does not qualify for data sharing because no datasets were created or examined for this investigation.

Declarations

Conflicts of interest The writers affirm that there is no possible conflict of interest between them.

Ethical approval We adhered to all relevant national and/or institutional norms for the use and care of animals.

Informed consent Formal consent is not required for this type of investigation.

References

Ariyo BO, Adesina LM, Ogunbiyi O (2025) Advanced microgrid systems: bridging the gap between renewable energy integration, stability, and scalable deployment—a review. J Eng Res Rev 2(2):105–105

Bahatti L, El Magri A, El Mezdi K (2024) An advanced algorithm for energy management and control for a multi-source microgrid that integrates electric vehicles with renewable energy. Eng Results 23:102642. https://doi.org/10.1016/j.rineng.2024.102642

Charadi S, Chakir HE, Redouane A, Akarne Y, El Hasnaoui A, Ettaoussi M (2025) Power flow management in hybrid AC/DC microgrids using the Artificial Bee Colony metaheuristic algorithm: a comparative study. Franklin Open. https://doi.org/10.1016/j.fraope.2025.100356

Dehghani M, Mahdavi M, Abdi A, Ghaffari M (2023) Coati optimization algorithm: a new bio-inspired metaheuristic algorithm for solving optimization problems. Knowl Based Syst 259:110011. https://doi.org/10.1016/j.knosys.2023.110011

Erdogan B, Tumay M, Savrun MM, Cuma MU (2025) A NARMA-L2 ANN-controlled hybrid storage-based energy management framework for enhanced microgrid operation. J Energy Storage 134:118241

Esparza A, Blondin M, Trovão JPF (2025) A review of optimization strategies for energy management in microgrids. Energies 18(13):3245

Güven AF (2024) Integrating electric vehicles into hybrid microgrids: a stochastic approach to future-ready renewable energy solutions and management. Energy. https://doi.org/10.1016/j.energy.2024.131968

Jose, A., &Seena Thomas, K. (2025). Unleashing the power of climate mitigation: Building a sustainable energy future through renewable energy, efficiency, and innovation. In: The intersection of global energy politics and climate change: A comprehensive analysis of energy markets and economics, Springer Nature Singapore, Singapore pp 299 319

Karthikeyan G, Selvakumar R, Prakash J (2024) Multi-source hybrid microgrid optimization using novel control algorithms. Renew Energy Syst 41(3):555–578

Krishnamoorthi K, Vidhya K (2024) A hybrid method for grid-connected photovoltaic interleaved inverters to improve power quality as much as possible. Energy Environ 35(1):244–274

Kumar RP, Karthikeyan G (2024) A multi-objective optimization solution for distributed generation energy management in microgrids with hybrid energy sources and battery storage system. J Energy Storage 75:109702. https://doi.org/10.1016/j.est.2024.109702

Mishra, D. P., Dash, A. K., Behera, S. R., Patro, K. A., &Salkuti, S. R. (2025). AI-driven optimization techniques for electric vehicle charging infrastructure. In: Artificial intelligence for integrated smart energy systems in *electric vehicles*. Springer Nature Switzerland Cham 283 308

Ntlhane RB, Bokopane L, Moyo N, Marumo MT (2024) Optimal power dispatching for a microgrid of grid-connected EV charging stations that includes peer-to-peer energy sharing, battery storage, and renewable energy. Energy Storage J 96:112435

Prajapati A, Paraye P, Ahirwar BK et al (2025) Tam CF. Artificial intelligence integration in solar-powered EV charging systems: challenges, opportunities, and future perspectives: a Prajapati. J Thermal Analy Calorim 150(16):12103–12134

Punitha, K., Aruna, R., &Kanthammal, S. (2024). Renewable-energy-powered EV charging station for microgrid: PSO-based controller for PV-powered EV charging station. In: Microgrids for commercial systems (pp. 381–408).

Rajesh CR, Alagan MT, Rajani B, Verma A (2025) Enhancing power quality in renewable energy systems through MAO-RERNN integrated harmonic mitigation. Energy. https://doi.org/10.1016/j.energy.2025.138296

Thangamuthu L, Anand R, Rajasekar S (2022) Design and development of extract maximum power from single-double diode PV model for different environmental conditions using BAT optimization algorithm. J Intell Fuzzy Syst 43(1):1091–1102

Wang P, Huang Z, Li J, Chen Y (2020a) Hybrid AC/DC microgrids with stochastic management that takes electric vehicle charging demands into account. Renew Energy 6:1338–1352

Wang S, Lu L, Han X, Ouyang M, Feng X (2020b) Virtual-battery-based droop control and energy storage system size optimization of a DC microgrid for electric vehicle fast charging station. Appl Energy 259:114146. https://doi.org/10.1016/j.apenergy.2019.114146

Xiao D, Deng W, Liu B, Huang W, Zhu Z (2025) Game-theoretic energy scheduling for community multi-energy system with electric vehicle aggregator leveraging quantum swarm intelligence. Int J Electr Power Energy Syst 172:111200

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.